Design and Optimization of a Hydraulic Oil Delivery System for Industrial Press Operation

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***Abstract*—This project presents the design and evaluation of a hydraulic piping system for delivering and returning oil to a hydraulic press while meeting specified performance metrics. The objective was to ensure 30 to 40 gpm delivery of oil at 50 psia, select an appropriate pump, and design a return system capable of operating under gravity while maintaining a maximum elevated tank volume of 10 gallons. The analysis applied conservation of mass and energy principles through the extended Bernoulli equation, along with Darcy-Weisbach and minor loss correlations, to determine system head requirements and flow behavior. A required pump head of approximately 107 ft at 40 gpm was calculated and used to select a suitable centrifugal pump that operates safely within its performance range and avoids cavitation. The return line was sized as a 2-inch schedule 40 PVC pipe, which was shown to provide sufficient flow under gravity without the need for an additional pump, and an overflow inlet was recommended to maintain consistent fluid levels and prevent air entrainment. A diaphragm valve was selected for effective flow control, and a cost analysis confirmed that the chosen configuration balances performance and economic efficiency. Overall, the results demonstrate that the system design is reliable, efficient, and well suited for the intended application.**

***Index Terms*—Hydraulic Press System, Pipe Sizing, Pump Sizing, Valve Specification**

I. Problem Statement

THE objective of this project is to analyze and design a proposed piping system that moves hydraulic oil from a supply tank to a manufacturing hydraulic press. The system must be engineered to reliably deliver oil at a pressure of 50 psia and maintain a minimum operating flow rate of 30 gpm, utilizing a control valve to restrict the flow rate if it exceeds 40 gpm. Furthermore, the project requires the sizing and selection of an appropriate pump, the sizing of the return line to ensure the elevated discharge tank retains no more than 10 gallons of oil, and the recommendation of optimal pipe diameters based on a minimum cost analysis.

The provided schematic from the problem statement shown in Fig. 1. illustrates the layout and flow path of the proposed fluid transport system. The diagram traces the movement of the hydraulic oil starting from the primary supply tank, moving through the pump and motor assembly, passing through a downstream control valve, and finally entering the hydraulic press. Additionally, the figure outlines the gravity driven return phase, showing where the used oil is discharged into an elevated tank on top of the press before flowing back down to the initial supply tank. Multiple measurements are providing in the schematic below including the total vertical distance from the fluid surface level in the supply tank to the fluid surface level in the elevated discharge tank, *H*, the vertical distance from the centerline of the pump inlet to the fluid surface level in the supply tank, *h*, and the vertical distance between the centerline of the pump's inlet piping and the centerline of its outlet piping, *d*. The values for these variables are shown in Table I below.

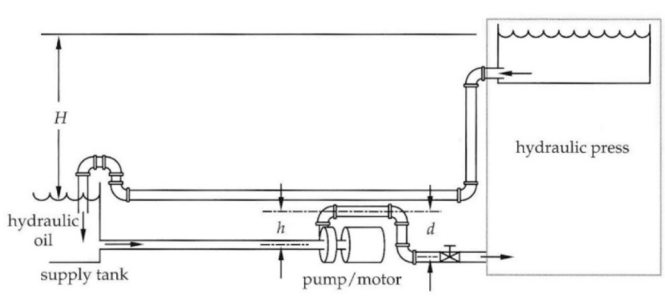

Fig. 1. Schematic representation of the proposed piping system detailing the oil supply route, return line, and pump placement

Table I
Key Elevation Parameters

Parameter	Value	Units
H	8	ft
h	10	in
d	12	in

To adequately size the pump, the required pump head was determined using the extended Bernoulli Equation, which applies conservation of energy to relate pressure velocity, pump energy input, and system losses in steady incompressible flow. Flow velocities in each pipe were found using the continuity equation based on conservation of mass. Energy losses were divided into major losses from pipe wall friction, calculated with the Darcy-Weisbach equation, and minor losses from fittings such as the entrance, elbows, valves, and exits, calculated using loss coefficients. To evaluate pipe friction, the Reynolds number was used to classify flow regime and determine the appropriate friction factor. Hydraulic oil density and viscosity were required as fluid properties because they affect pressure head conversion and flow resistance. By combining pressure rise requirements, elevation change, velocity differences, and total head losses, the total pump head required to deliver the hydraulic oil at the specified flow rate and pressure was obtained.

II. Procedure

Part a: Pump Sizing

To determine the size of the pump to move hydraulic oil from the supply tank to the press, the pump head required had to first be calculated. Multiple parameters from the problem statement were needed to solve for the pump head and are included in Table II below where Q is the flow rate of the oil, P_1 is the atmospheric pressure, P_2 is the pressure of the hydraulic oil, L_1 is the length of the inlet line pipe and L_2 is the length of the outlet line pipe.

Table II
Flow Rate, Pressure and Length Parameters

Parameter	Value	Units
Q	40	gpm
P_1	14.7	psia
P_2	50	psia
L_1	2.5	m
L_2	6	m

The first step was to calculate the flow velocity in each pipe. Before this could be done, flow rate had to be converted to metric units, pipe diameters were found from a commercial industrial supply PVC chart [1], and area was solved for the inlet pipe, A_1, and the outlet pipe, A_2, using (1) below. Then the flow velocity of the inner pipe, V_1, and the flow velocity of the outer pipe, V_2, were determined using (2).

$$Q = 40\ \cancel{gpm} * \left(6.309\ \times 10^{-5} \frac{m^3/s}{\cancel{gpm}}\right) = 0.00252\ m^3/s$$

$$D_1 = 2.067\ in = 0.0525\ m$$

$$D_2 = 1.61\ in = 0.0409\ m$$

$$A = \frac{\pi D^2}{4}\ Ex.8.11, p.277\ Fox\ and\ McDonald\ [2] \tag{1}$$

$$A_1 = \frac{\pi {D_1}^2}{4} = \frac{\pi(0.0525\ m)^2}{4} = 0.00216\ m^2$$

$$A_2 = \frac{\pi {D_2}^2}{4} = \frac{\pi(0.0409\ m)^2}{4} = 0.00131\ m^2$$

$$V = \frac{Q}{A}\ Eqn.4.14b, p.78\ Fox\ and\ McDonald\ [2] \tag{2}$$

$$V_1 = \frac{Q}{A_1} = \frac{0.00252\ \frac{m^{\cancel{3}}}{s}}{0.00216\ \cancel{m^2}} = 1.166\ m/s$$

$$V_2 = \frac{Q}{A_2} = \frac{0.00252\ \frac{m^{\cancel{3}}}{s}}{0.00131\ \cancel{m^2}} = 1.921\ m/s$$

Next, the flow velocities previously solved for each pipe were equated with the density, ρ, and kinematic velocity, v, of the oil using (3). In finding density and kinematic velocity, the oil was assumed to be ISO VG 32 hydraulic oil at 40 °C [3].

$$\rho = 857\ kg/m^3$$

$$v = 3.2\ \times 10^{-5}\ m^2/s$$

$$Re = \frac{VD}{v}\ Eqn.7.11, p.212\ Fox\ and\ McDonald\ [2] \tag{3}$$

$$Re_1 = \frac{V_1 D_1}{v} = \frac{\left(1.166\ \frac{\cancel{m}}{\cancel{s}}\right)(0.0525\ \cancel{m})}{3.2\ \times 10^{-5}\ \cancel{m^2/s}} = 1912.59$$

$$Re_2 = \frac{V_2 D_2}{v} = \frac{\left(1.921\ \frac{\cancel{m}}{\cancel{s}}\right)(0.0409\ \cancel{m})}{3.2\ \times 10^{-5}\ \cancel{m^2/s}} = 2455.03$$

The flow of the inner pipe was deemed laminar due to the Reynolds number, Re_1, being less than 2300, while the flow for the outer pipe was between transitional and turbulent due to the Reynolds number, Re_2, being greater than 2300 by a relatively small margin. Because of this difference in flow behavior between the inner and outer pipes, two different methods of calculating friction factors were employed. For the inner pipe, friction factor, f_1, was solved with a simple equation involving Reynolds number shown in (4), while the friction factor for the outer pipe, f_2, was solved using (5) which involved a slightly more complex relation with Reynolds number.

$$f_1 = \frac{64}{Re_1}\ Eqn.8.36, p.256\ Fox\ and\ McDonald\ [2] \tag{4}$$

$$f_2 = \frac{0.316}{Re_2^{1/4}}\ Eqn.8.38, p.257\ Fox\ and\ McDonald\ [2] \tag{5}$$

$$f_1 = \frac{64}{Re_1} = \frac{64}{1912.59} = 0.0335$$

$$f_2 = \frac{0.316}{(2455.03)^{1/4}} = 0.0449$$

All parameters needed to find major head loss were now determined, but first the friction head loss for each pipe had to be found using (6). Then, the major head loss was found by adding the friction head loss of the inner pipe, h_{f1}, and the friction head loss of the outer pipe, h_{f2}, as shown in (7).

$$h_f = f\left(\frac{L}{D}\right)\left(\frac{V^2}{2g}\right) Eqn.\,8.35, p.\,255\ Fox\ and\ McDonald\ [a] \tag{6}$$

$$h_{f1} = f_1\left(\frac{L_1}{D_1}\right)\left(\frac{V_1^2}{2g}\right) = (0.0335)\left(\frac{2.5\ \cancel{m}}{0.0525\ \cancel{m}}\right)\left(\frac{\left(1.166\frac{m}{s}\right)^{\cancel{2}}}{2\left(9.81\frac{\cancel{m}}{\cancel{s^2}}\right)}\right) = 0.111\ m$$

$$h_{f2} = f_2\left(\frac{L_2}{D_2}\right)\left(\frac{V_1^2}{2g}\right) = (0.0449)\left(\frac{6\ \cancel{m}}{0.0409\ \cancel{m}}\right)\left(\frac{\left(1.921\frac{m}{s}\right)^{\cancel{2}}}{2\left(9.81\frac{\cancel{m}}{\cancel{s^2}}\right)}\right) = 1.238\ m$$

$$h_{major} = h_{f1} + h_{f2}\ Eqn.\,8.29, p.\,263\ Fox\ and\ McDonald\ [2] \tag{7}$$

$$h_{major} = h_{f1} + h_{f2} = 0.111\ m + 1.238\ m = 1.349\ m$$

In addition to major head loss, minor head loss had to be considered before calculating total head loss. Minor loss coefficients were found based off the geometry and set up of the apparatus including the entrance, three elbows, fully open valve and exit [4]. The minor loss coefficient for the entrance, $K_{entrance}$, elbows, K_{elbow}, fully open valve, K_{valve}, and the exit, K_{exit}, are shown in Table III below and then the total minor loss coefficient, K_{total}, was solved using (8). Finally, minor head loss, h_{minor}, was determined by equating the total minor loss coefficient with the velocity of the outer pipe and the gravitational constant as shown in (9).

Table III
Minor Head Loss Coefficients

Coefficient	Value (m)	Quantity
$K_{entrance}$	0.5	1
K_{elbow}	0.3	3
K_{valve}	10.0	1
K_{exit}	1.0	1

$$K_{total} = K_{entrance} + 3K_{elbow} + K_{valve} + K_{exit} \tag{8}$$

$$K_{total} = 0.5 + 3 * 0.3 + 10.0 + 1 = 12.4$$

$$h_{minor} = K_{total}\left(\frac{V_2^2}{2g}\right) Eqn.\,8.40, p.\,258\ Fox\ and\ McDonald\ [2] \tag{9}$$

$$h_{minor} = (12.4)\left(\frac{\left(1.921\frac{m}{s}\right)^{\cancel{2}}}{2\left(9.81\frac{\cancel{m}}{\cancel{s^2}}\right)}\right) = 2.332\ m$$

With major and minor head loss now determined, total head loss, h_L, was solved by adding these two values as shown in (10). The total head loss was then input to the extended Bernoulli's equation shown in (11), where the change in height between the inlet and outlet, $z_2 - z_1$, was assumed to be -0.0508 m, to solve for the required pump head, H_p.

$$h_L = h_{major} + h_{minor}\ Eqn.8.29, p.263\ Fox\ and\ McDonald\ [2] \tag{10}$$

$$h_L = h_{major} + h_{minor} = 1.349\ m + 2.332\ m = 3.681\ m$$

$$H_p = \frac{P_2 - P_1}{\rho g} + \frac{V_2^2 - V_1^2}{2g} + (z_2 - z_1) + h_L\ Eqn.8.30, p.253\ Fox\ and\ McDonald\ [2] \tag{11}$$

$$(P_2 - P_1) = (50\ psia - 14.7\ psia) = 35.3\ \cancel{psia} * \frac{6894.76\ Pa}{1\ \cancel{psia}} = 243385\ Pa$$

$$(z_2 - z_1) = -0.0508\ m$$

$$H_p = \frac{P_2 - P_1}{\rho g} + \frac{V_2^2 - V_1^2}{2g} + (z_2 - z_1) + h_L$$

$$H_p = \frac{243385\ \cancel{Pa}}{\left(857\frac{\cancel{kg}}{m^{\cancel{3}}}\right)\left(9.81\frac{\cancel{m}}{\cancel{s^2}}\right)} + \frac{\left(1.921\frac{m}{\cancel{s}}\right)^{\cancel{2}} - \left(1.166\frac{m}{\cancel{s}}\right)^{\cancel{2}}}{2\left(9.81\frac{\cancel{m}}{\cancel{s^2}}\right)} + (-0.0508\ m) + 3.681\ m$$

$$H_p = 32.698\ \cancel{m} * \frac{3.281\ ft}{1\ \cancel{m}} = 107.283\ ft$$

$$\therefore H_p = 107.283\ ft$$

The final calculated required pump head was 107.283 ft which means that the pump needed for this apparatus must be able to lift the hydraulic oil to a minimum height of 107.283 ft. With the required pump head determined, and the known flow rate of 40 gpm from the problem statement, a suitable pump could now be determined from a manufacturing website [5]. The type of pump selected was an end-suction e-1510 centrifugal pump because its operating characteristics align well with the system's flow, head and fluid requirements. To properly size the pump, the calculated pump head and provided maximum flow rate were input along with the default settings and number of pumps equal to one. The result was an end-section e-1510 1.25 AD size centrifugal pump. This specific size provided the highest efficiency, 46.7%, with the lowest required motor power, 5 hp, highlighting it as the best choice for this application.

Part b: Return Line Sizing

The return line needed to be sized so that oil in the tank would not exceed 10 gallons and then it had to be decided whether to modify the inlet in the return line to make it an overflow inlet. Similar parameters and properties were considered as the pervious calculations for pump head, and the length of the return line, L, was equal to 12 meters or 39.37 ft. First, the maximum depth of oil in the tank, h_{oil}, had to be calculated by equating the cross-sectional area of the tank, A_c, with the maximum volume of oil, V_{max}, as shown in (12) below.

$$A_c = 2.5\ ft^2$$

$$V_{max} = 10\ \cancel{gal} * \frac{0.1337\ ft^3}{1\ \cancel{gal}} = 1.337\ ft^3$$

$$h_{oil} = \frac{V_{max}}{A_c}\ Ex.2.3, p.31\ Fox\ and\ McDonald\ [2] \tag{12}$$

$$h_{oil} = \frac{V_{max}}{A_c} = \frac{1.337\ ft^3}{2.5\ ft^2} = 0.535\ ft$$

Next, the available head due to elevation change, H_a, was calculated using (13) which depended on the previously solved oil depth and the total vertical distance from the fluid surface level in the supply tank to the fluid surface level in the elevated discharge tank, H.

$$H_a = H + h_{oil} \tag{13}$$

$$H_a = H + h_{oil} = 8\ ft + 0.535\ ft = 8.535\ ft$$

Next, the two different pipe sizes, 2-nominal and 1.5-nominal schedule 40 PVC had to be analytically compared to determine which size resulted in head loss less than available head due to elevation change. First, velocity was calculated for the 2-inch pipe, V_2, and the 1.5-inch pipe, $V_{1.5}$, using (2).

$$V_2 = \frac{Q}{A_2} = \frac{0.0891\ \frac{ft^{\cancel{3}}}{s}}{0.0233\ \cancel{ft^2}} = 3.83\ ft/s$$

$$V_{1.5} = \frac{Q}{A_{1.5}} = \frac{0.0891\ \frac{ft^{\cancel{3}}}{s}}{0.0141\ \cancel{ft^2}} = 6.23\ ft/s$$

Reynolds number for the 2-inch and 1.5-inch pipe, Re_2 and $Re_{1.5}$, were then calculated using (3) to determine the flow regime within each pipe.

$$Re_2 = \frac{V_2 D_2}{v} = \frac{\left(3.83\ \frac{\cancel{ft}}{\cancel{s}}\right)(0.1722\ \cancel{ft})}{1.07\ \times 10^{-3}\ \cancel{ft^2/s}} = 616$$

$$Re_{1.5} = \frac{V_{1.5} D_{1.5}}{v} = \frac{\left(6.23\ \frac{\cancel{ft}}{\cancel{s}}\right)(0.1342\ \cancel{ft})}{1.07\ \times 10^{-3}\ \cancel{ft^2/s}} = 781$$

Since the Reynold's number was less than 2300 for both pipe sizes, the flow regime was deemed laminar for both and thus the friction factor for the 2-inch and 1.5-inch pipe, f_2 and $f_{1.5}$, were solved using (4).

$$f_2 = \frac{64}{Re_2} = \frac{64}{616} = 0.104$$

$$f_{1.5} = \frac{64}{Re_{1.5}} = \frac{64}{781} = 0.0819$$

Finally, major head loss, minor head loss, and total head loss for the 2-inch and 1.5-inch pipe was solved using (6), (9), and (10), respectively. For the minor head loss, a loss coefficient, K, of 6.3 was used based off the entrance, elbows and exit geometry of the return line.

$$h_{2major} = f_2\left(\frac{L}{D_2}\right)\left(\frac{V_2^2}{2g}\right) = (0.104)\left(\frac{39.37\ \cancel{ft}}{0.1722\cancel{ft}}\right)\left(\frac{\left(3.83\frac{ft}{s}\right)^{\cancel{2}}}{2\left(32.2\ \frac{\cancel{ft}}{\cancel{s^2}}\right)}\right) = 5.4\ ft$$

$$h_{1.5major} = f_{1.5}\left(\frac{L}{D_{1.5}}\right)\left(\frac{V_{1.5}^2}{2g}\right) = (0.0819)\left(\frac{39.37\,\cancel{ft}}{0.1342\,\cancel{ft}}\right)\left(\frac{\left(6.23\frac{ft}{s}\right)^{\cancel{2}}}{2\left(32.2\,\frac{\cancel{ft}}{\cancel{s^2}}\right)}\right) = 14.5\,ft$$

$$h_{2mino} = (6.3)\left(\frac{\left(3.83\frac{ft}{s}\right)^{\cancel{2}}}{2\left(32.2\,\frac{\cancel{ft}}{\cancel{s^2}}\right)}\right) = 1.44\,ft$$

$$h_{1.5minor} = (6.3)\left(\frac{\left(6.23\frac{ft}{s}\right)^{\cancel{2}}}{2\left(32.2\,\frac{\cancel{ft}}{\cancel{s^2}}\right)}\right) = 3.8\,ft$$

$$h_{2L} = h_{2major} + h_{2minor} = 5.4\,ft + 1.44\,ft = 6.42\,ft$$

$$h_{1.5L} = h_{1.5major} + h_{1.5minor} = 14.5\,ft + 3.8\,ft = 18.3\,ft$$

Comparing the total head loss for each pipe to the available head, H_a, the head loss for the 2-inch pipe is less than the available head while the head loss for the 1.5-inch pipe is greater than the available head. Therefore, the 2-inch pipe was selected for the return line for optimal drainage without overflow caused by greater head loss than available head.

III. Results

Part a: Results

In part a, the required pump head, H_p, was calculated to determine what size pump to select for the operation. Table IV shows the results of this calculation including important intermediate variables such as friction head loss in the inner and out pump, h_{f1} and h_{f2}, major head loss, h_{major}, minor head loss, h_{minor}, and total head loss, h_L, which were needed to solve for pump head.

Table IV
Head Calculation Results

Parameter	Value	Units
h_{f1}	0.1104	m
h_{f2}	1.2384	m
h_{major}	1.3488	m
h_{minor}	2.3318	m
H_p	107.28	ft

Once the pump head was solved, it was input to a pump manufacturing website along with the flow rate of 40 gpm to select a properly sized pump for the application. The result was an end-suction e-1510 1.25AD centrifugal pump as shown in Fig. 2. below with an efficiency of 46.7% and a motor power of 5 hp. Additional metrics of the selected pump include a discharge size of 1.25 inches, discharge velocity of 8.6 feet-per-second, suction size of 1.5 inches, suction velocity of 6.4 feet-per-second, weight of 130 lbs, impeller diameter of 5.125 inches and floor space of 3.4 square feet.

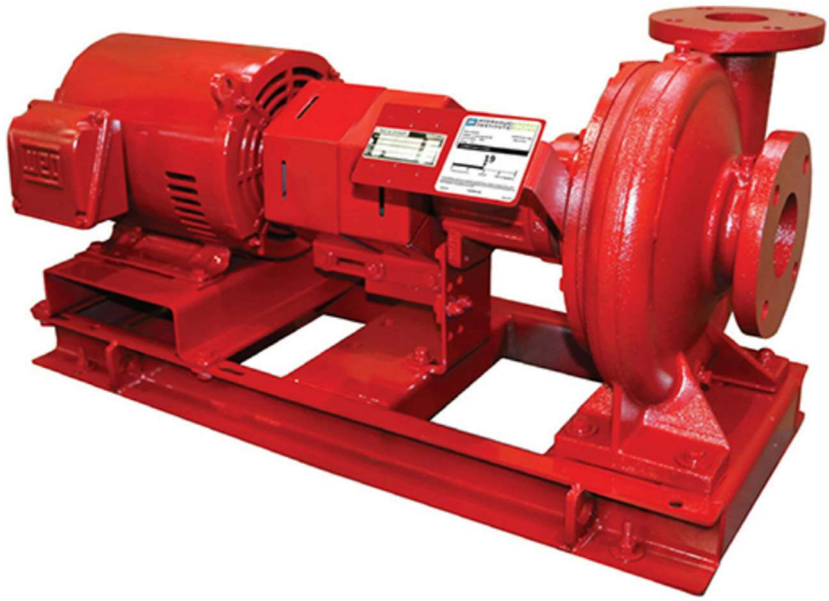

Fig. 2. End-suction e-1510 1.25AD centrifugal pump selected for the hydraulic oil operation

The pump selection was further verified by the corresponding performance curve, shown in Fig. 3. below, that illustrates the relationship between flow rate and head produced by the pump, along with the efficiency, power requirements, and operating limits. The main curve shows that as flow rate increases, the head decreases, reflecting the pump's energy distribution characteristics. The intersection of the system curve with the pump curve defines the actual operating point, which in this case occurs near 40-41 gpm and approximately 110 to 113 ft of head.

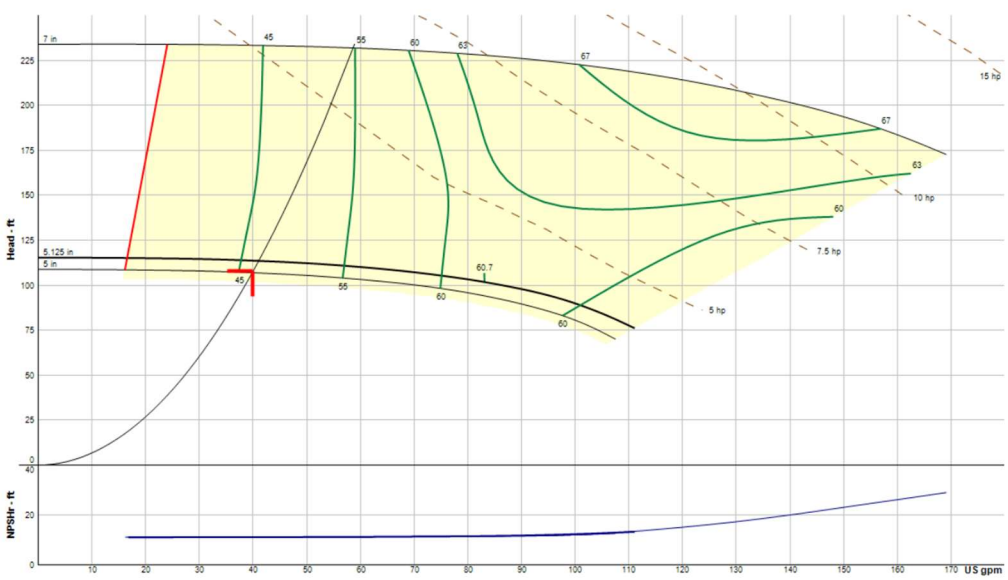

Fig. 3. Performance curve of selected pump for hydraulic oil application

Part b: Return Line Sizing

The 2-inch-nominal schedule 40 PVC pipe was selected for the return size since the total head loss was less than the available head. As for the necessity of a modified inlet, this was deemed a good idea to prevent air from entering

the return line and avoid the 10-gallon volume limit from being exceeded. The oil level must rise slightly to enter the inlet, so the position should be set less than the 10-gallon line to ensure that this maximum volume is not exceeded in operation. Hence, the overflow inlet is a good idea to minimize inconsistency in the hydraulic fluid in the system and increase lifespan.

Part c: Drainage Evaluation

The return line conveys oil fast enough because with the 2-nominal pipe size being used, less resistance is created than the pressure provided by gravity (6.42 ft < 8.535 ft). Additionally, a pump is not necessary in the return line because the potential energy from the elevated tank is more than enough to overcome the friction of the return line. Adding a pump would only increase operational complexity and cost.

Part d: Valve Specification

The valve determined to be the best and most applicable for this system is a ½'' Hayward DAB Series True Union CPVC Diaphragm Valve shown in Fig. 4. below [7]. This specific valve can control the flow by adjusting its opening and can specifically reduce the flow if the 40 gpm limit is exceeded. It can also be implemented into a PVC piping system because it is available with socket connections that can be glued to the PVC pipe. The only issue is that this valve would need to be integrated into a 1.25'' pipe size. This can easily be solved by implementing a 1.25'' x ½'' PVC reducing bushing to be able to connect the valve to the pipe. This valve also adjusts flow gradually and not abruptly, which can prevent a sudden change in pressure and is in fact good for protecting the hydraulic press. The pressure of the hydraulic oil delivered to the press is also required to be at 50 psia [5]. This valve is rated for much higher pressures (up to 150 psi), so it fits well within that range. A diaphragm valve is a type of control valve which is especially suitable for this type of application which requires flow to be adjusted. Additionally, this valve contains o-rings to prevent leaks and it works well with viscous fluids such as oil.

Fig. 4. ½" Hayward DAB Series True Union Diaphragm Valve selected for hydraulic oil operation

Part e: Minimum Cost Analysis

A cost analysis was performed comparing the selected 1.25 in, 5 hp pump [5] to two others. One of them was 1.5 in, 7 hp pump [5], and the other was a 2 in, 7 hp pump [5]. Energy cost was estimated, which is the amount of money it costs to run the pump using electricity. Energy cost, is dependent on power, the amount of hours the pipe runs for, and the electricity rate. Assuming the pump runs for 8 hours a day, that would account for 2,920 hours per year.

Multiplying each of the pumps BHP value by 0.746 will solve for power. Finally, an electricity rate is assumed to be $0.1745 $/kWh, which is standard in The United States [6]. Multiplying the hours worked per year, the power, and the operating cost by each other, the cost can be determined (Table V). Additionally, assuming each pump has a life span of 10 years, the total cost for its entire lifespan can be calculated by multiplying the annual cost by 10.

Table V
Minimum Cost Analysis Results

Pump Size (in)	Efficiency (%)	Motor Power (hp)	Annual Energy Cost ($)	Lifetime Cost ($)
1.25	46.7%	5	$653	$6,530
1.5	42.3%	7	$669	$6,690
2	39%	7	$739	$7,390

As seen in the data, the selected pump that has a discharge size of 1.25 in is the most cost-effective by costing the least amount of money annually. Furthermore, the motor power is lowest for the selected pump, and a higher value means that the pump requires more power from the motor which can cause higher forces on the pump components inevitably creating wear. This can lead to more maintenance, which will greatly add to the expenses. Therefore, a lower motor power is more ideal for minimizing costs as well. The 1.25 in pump also has the highest efficiency which means that less energy is wasted, so the pump uses less power to perform the same function which lowers the operating cost. This cost analysis verifies that the 1.25 in, 5 hp pump is a very cost-effective option.

IV. Discussion

The objective of this analysis was to design and evaluate a hydraulic piping system capable of reliably supplying hydraulic oil to a press while meeting specified performance requirements. The system must deliver between 30 to 40 gpm of oil at a pressure of 50 psia, requiring proper pump sizing and selection based on calculated energy demands. In addition, the return line must be designed to safely transport oil back to the supply tank under gravity while maintaining acceptable fluid levels in the elevated tank. The analysis also includes selecting appropriate components, such as pipe diameters and a control valve, to ensure efficient, safe and controllable system operation.

The calculated pump head is considered reliable because it is grounded in fundamental fluid mechanics principles and aligns closely with the actual performance of the selected pump. The extended Bernoulli equation was used to account for all energy contributions in the system, including pressure requirements, elevation changes, velocity effects, and both major and minor losses, with conservative assumptions such as designing for the maximum flow rate of 40 gpm. This resulted in a required pump head of approximately 107 ft, which closely matches the pump's operating point of about 41 gpm at 113 ft. This near agreement confirms that the analytical model accurately represents the system. Additionally, the pump operates within a stable and efficient region of its performance curve, avoiding undesirable conditions such as runout or shutoff. The selected pump also satisfies cavitation constraints, as the available NPSH significantly exceeds the required NPSH. Together, the strong agreement between calculated requirements and pump performance, proper operating region placement, and safe suction conditions demonstrate that the pump selection is both valid and well suited for the application.

The results for the remaining design components further support that the system is both feasible and well optimized. For the return line sizing, selecting a 2-inch-nominal schedule 40 PVC pipe is justified because the calculated head losses are lower than the available gravitational head, ensuring reliable flow while providing a safety margin. The recommendation of an overflow inlet is also appropriate as it maintains a consistent fluid level, prevents air entrainment, and ensures the 10-gallon volume limit is not exceeded, all of which improve system stability and longevity. The drainage evaluation confirms that gravity alone is sufficient to drive the return flow, since the available head exceeds total losses, eliminating the need for an additional pump and thereby reducing cost and complexity. The diaphragm valve selection is also suitable, as it allows for smooth and precise flow control, operates effectively with viscous fluids, and is rated well above system pressure, ensuring safe operation. Finally, the minimum cost analysis supports the chosen configuration by demonstrating that the selected pump provides adequate performance at a lower operating cost compared to larger alternatives, balancing both initial and long-term expenses. Collectively, these results validate that each component of the system has been selected to meet performance requirements while maintaining efficiency, safety, and cost-effectiveness.

Conclusion

Overall, the hydraulic system design meets all required performance and operational criteria through a combination of sound analysis and appropriate component selection. The pump was accurately sized and verified to deliver the necessary flow and head while operating safely and efficiently. The return line was properly designed to allow gravity-driven flow without the need for additional pumping, and the inclusion of an overflow inlet improves system reliability and prevents air entrainment. The selected control valve provides effective flow regulation while maintaining compatibility with system pressures and fluid properties. Finally, the cost analysis confirms that the chosen components achieve a balance between performance and economic efficiency. Together, these results demonstrate that the system is well designed, practical, and capable of long-term operation.

V. References

[1] M. Ligon, "PVC Piping Dimension Chart for Sch 40 & Sch 80," *Commercial Industrial Supply*, Jun. 02, 2025. https://www.commercial-industrial-supply.com/blogs/resource-center/pvc-pipe-and-fittings-dimensions

[2] R. W. Fox and J. W. Mitchell, Fox and McDonald's Introduction to Fluid Mechanics. Hoboken: Wiley, 2019.

[3] "ISO Grade Oils - Viscosities and Densities," *www.engineeringtoolbox.com*. https://www.engineeringtoolbox.com/iso-grade-oil-d_1207.html

[4] Engineering Toolbox, "Minor or Dynamic Loss Coefficients for Pipe or Tube System Components," *Engineeringtoolbox.com*, 2019. https://www.engineeringtoolbox.com/minor-loss-coefficients-pipes-d_626.html

[5] "ESP Systemwize," *www.esp-systemwize.com*. https://www.esp-systemwize.com/pumps/details

[6] Choose Energy, "Electricity Rates by State (February 2020) | ChooseEnergy.com®," *@ChooseEnergy*, 2020. https://www.chooseenergy.com/electricity-rates-by-state/

[7] "1/2" Hayward DAB Series True Union CPVC Diaphragm Valve," *Pvcpipesupplies.com*, 2018. https://pvcpipesupplies.com/12in-hayward-150psi-fpm-socket-threaded-trueunion-cpvc-diaphragm-valve-dab2005upf.html (accessed Apr. 15, 2026).

[8] K. Schulze, *EML4147C Thermal Sciences design project Fall 25.docx*.Gainesville, FL: 2026